

Amoebiasis (Amoebic Dysentery)

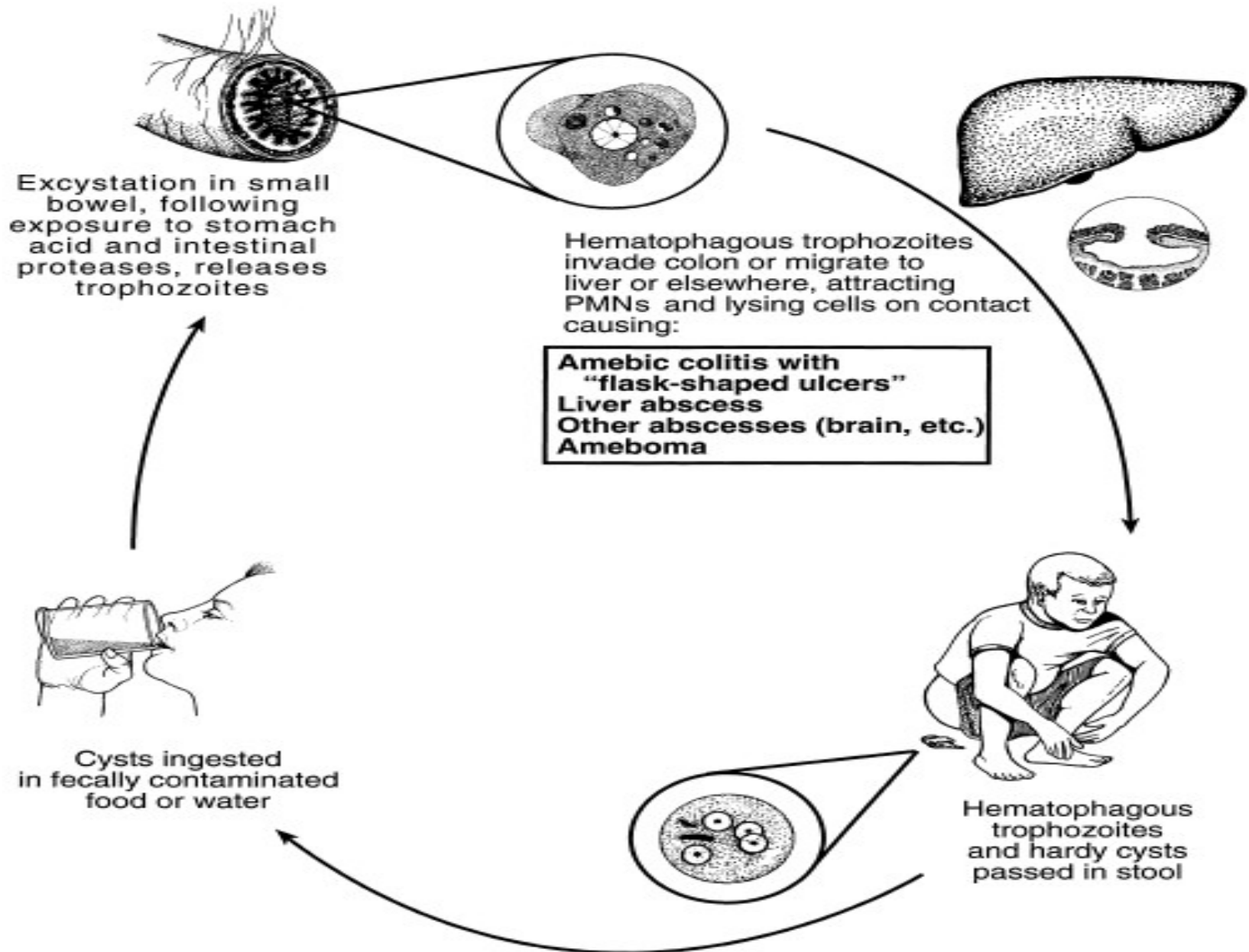
- **Definition**
- An infection caused by *Entamoeba histolytica*, which is commonly asymptomatic but may produce clinical manifestations ranging from mild diarrhea to severe dysentery sometimes reaching the liver and other extra-intestinal organs and causing abscesses, with relatively high case fatality in untreated cases.
- **Infectious agent**
 - . *Entamoeba histolytica*

Epidemiology

- **Occurrence**- worldwide but most common in the tropics and sub-tropics.
- Prevalent in areas with poor sanitation, in mental institutions and homosexuals.
- Invasive amoebiasis is mostly a disease of young people (adults).
- **Mode of transmission** – Fecal-oral transmission by ingestion of food or water contaminated by feces containing the cyst.
- Acute amoebic dysentery poses limited danger.

- **Incubation period**- Variable from few days to several months or years; commonly 2-4 weeks.
- **Period of communicability**- During the period of passing cysts of *E. histolytica*, which may continue for years.
- **Susceptibility and resistance**- Susceptibility is general. Susceptibility to reinfection has been demonstrated .

Life cycle



Clinical Manifestation

- **NON-INVASIVE**
- Starts with a prodromal episode of diarrhea, abdominal cramps, nausea, vomiting and tenesmus.
- With dysentery, feces are generally watery, containing mucus and blood

- **INVASIVE**

- ✓ The non-invasive infection can persist or progress to an invasive by
 - Penetration of mucus layer
 - Contact-dependent killing of epithelial cells
 - Breakdown of tissues (extracellular matrix)
 - Contact-dependent killing of neutrophils, leukocytes, etc

Pathogenesis

- Virulence factors: lectin, cysteine protease & amoebapore
- 1st stage invasive starts when adhesion of colonic mucins via parasite surface Gal/GalNAc lectin.
- Human colonic mucin layer of large intestine may be the first receptor encountered by lectin Gal/GalNAc lectin is associated with complement resistance to trophozoite.
- Activation of parasite motility is a prerequisite for invasion

- The hallmark of amebiasis include:
 - degradation of extracellular matrix by amebic protease
 - production of toxic factors
 - activation of host cells from host immune system
 - killing & phagocytosis of human cells
- It induces apoptosis & phagocytosis of human target cells

Mechanisms of Cytotoxicity

- At least **5 mechanisms** explain pathogenicity *E. histolytica*

1. **Direct contacts** with the host tissues.

e.g: adherence to host cell by release of adhesin molecules

2. Release of soluble **toxic** metabolites.

e.g: **amoebapore** are capable of forming a hole in target cell

3. High **enzymatic** activities of the amoebas

e.g: **enzymes** released by parasite damaging the host cell.

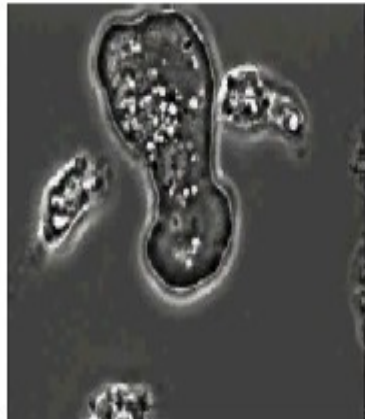
4. Interfering with the immune response of the host.

eg: Antigens released by the amoeba cell can lead to:

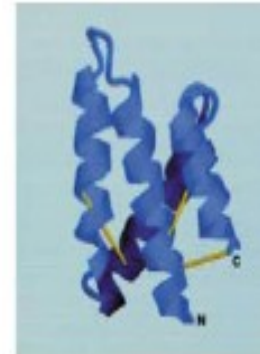
- inflammatory reaction
- allergies
- immunodepression
- over stimulation of cytokines
- formation of immune complexes
- formation of autoantibodies

5. *E. histolytica* induces apoptosis of the host cells during tissue invasion

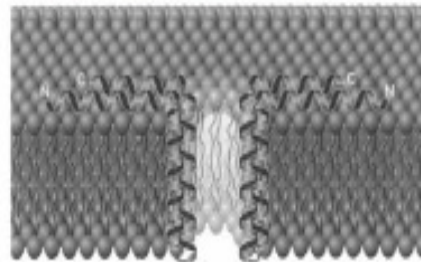
Pathogenic amoeba show contact dependent killing



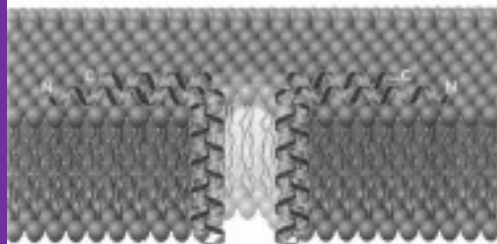
Amoebapores one of the candidate pathogenicity factors



- Family of small (77 AA) proteins contained in secretory granules
- Similar in structure and function to NK lysins
- Used to kill bacteria and host cells
- Amoebapores insert into target membranes and form ion channels
- Amoeba mutants which make less amoebapores cause less disease in animal model studies

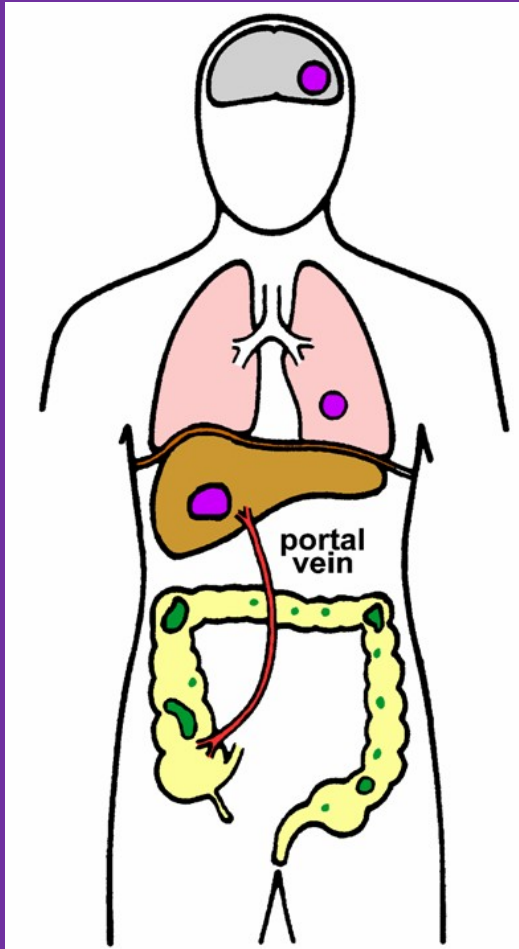


Toxins -- Amoebapores



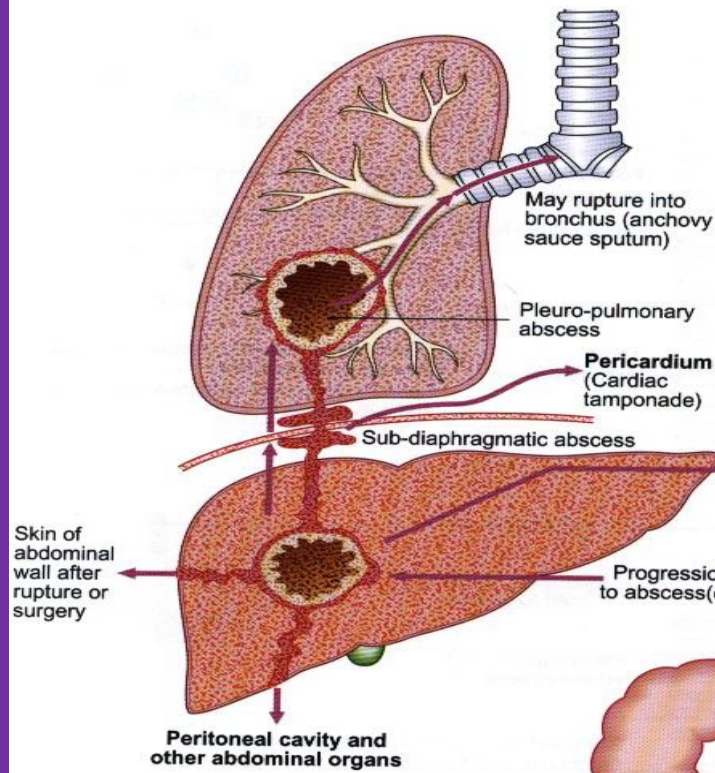
- Amoebapores insert into target membranes and form ion channels

Intestinal amobiasis can lead to secondary infection and extra intestinal amobiasis

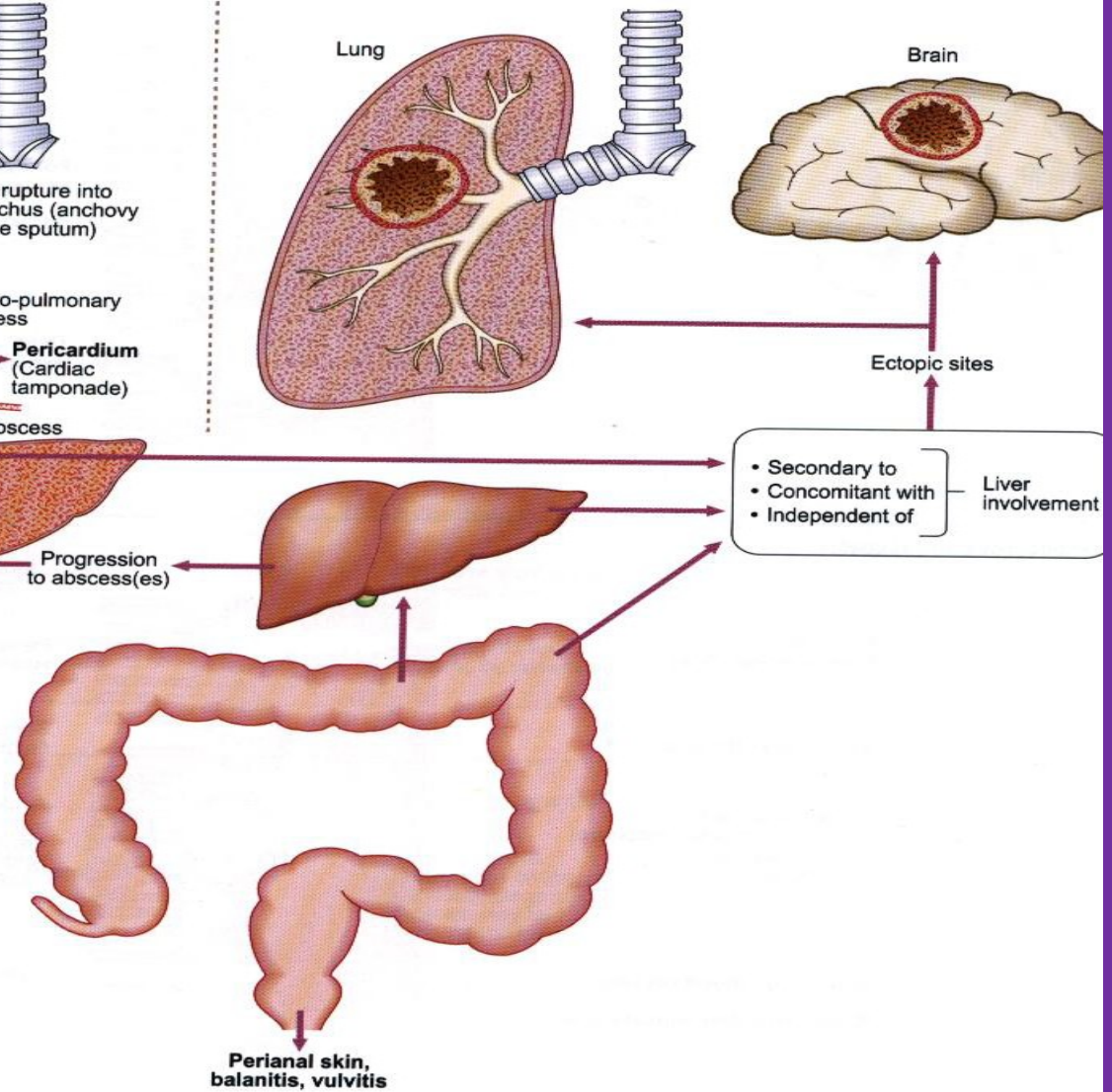


- Metastasis via blood stream
- Primarily liver (portal vein)
 - Other sites less frequent
- Amoeba-free stools common
- High antibody titers

Direct extension



Haematogenous spread



Amoebic liver abscess



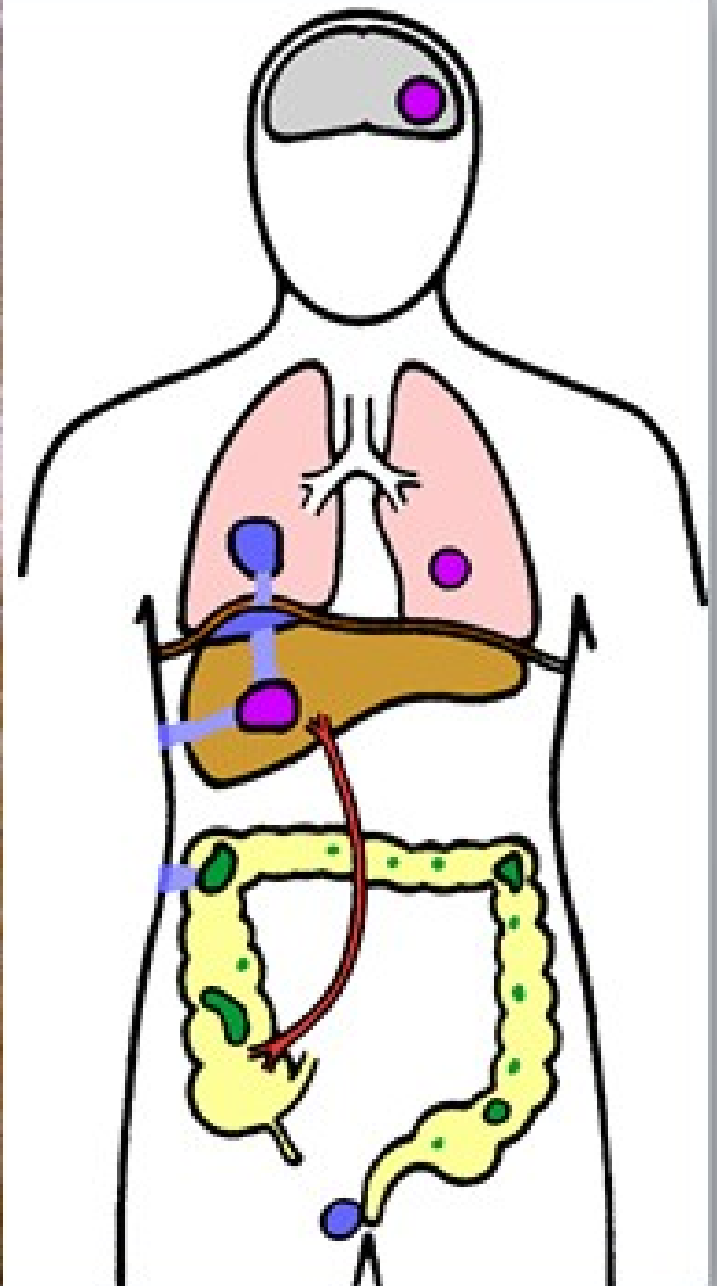
- Most common form of extraintestinal amebiasis
- symptoms are right upper quadrant pain and fever
- Fast growing abscess filled with:
 - chocolate-colored 'pus'
 - necrotic material
 - usually bacteria free
 - amoebae are found only at borders



- 30-50% of patients with liver abscess show also pneumonic involvement
- Rupture is again a major threat, especially rupture into the pericardium
- Draining abscesses is today only performed in extreme cases when rupture is feared

Pulmonary Amoebiasis

- rarely primary
- rupture of liver abscess through diaphragm
- 2° bacterial infections common
- fever, cough, dyspnea, pain



- Cutaneous Amoebiasis
- intestinal or hepatic fistula
- mucosa bathed in fluids containing trophozoites
 - perianal ulcers
 - urogenital (eg, labia, vagina, penis)



- **Diagnosis**

- Demonstration of *Entamoeba histolytica* cyst or trophozoite in stool for intestinal amebiasis
- demonstration of the trophozoites from Aspiration or other biopsy sample.

Treatment

- ❑ **Invasive intestinal amoebiasis:** Tissue amoebicides such as nitroimidazoles (Metronidazole or Tinidazole) combined with luminal amoebicides.
- Metronidazole
 - - Adult- 750 mg po TID for 10 days

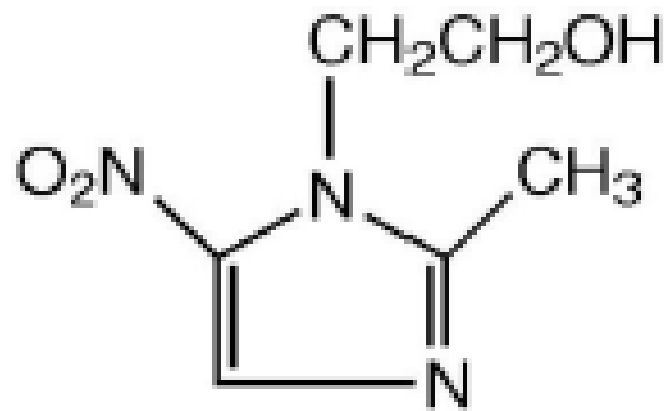
-Child- 15 mg/kg every 8 hours maximum of 750
mg po TID for 10 days

➤ Tinidazole

- Adult- 1gm po BID for 3 days
- Child-50 mg/kg maximum of 2 gm daily for 3
days

❑ Amebic liver abscess is best treated with metronidazole.

Metronidazole is the drug of choice for amebiasis



- Several drugs are available to clear symptomatic and asymptomatic enteric (luminal) infection (e.g. dichloroacetamides which have unknown mode of action)
- Metronidazole (Flagyl) is the drug of choice for invasive amoebiasis (and should be combined with a lumen acting drug as it is not fully effective on luminal stages)
- Metronidazole is a prodrug which is activated by an enzyme involved in the microaerobic fermentation metabolism of *E. histolytica*

Prevention and control

1. Adequate treatment of cases
2. Provision of safe drinking water
3. Proper disposal of human excreta (feces) and
Hand washing following defecation.
4. Cleaning and cooking of local foods (e.g. raw
vegetables)
to avoid eating food contaminated with feces.

Pathogenic Free-Living Amebae

Outline

Naegleria fowleri and *Acanthamoeba* spp

- General feature of pathogenic free living amoeba
- Epidemiology
- Morphology
- Transmission and life cycle
- Pathology and clinical manifestation
- Laboratory diagnosis

Free Living pathogenic Amoeba

- Survive and replicate in the environment without a host (no insect vectors etc.)
- Infection of humans is **infrequent** but life threatening
- 3 genera which can cause disease in humans:
 - *Naegleria*- Primary amebic meningoencephalitis (PAM)
 - *Acanthamoeba* -granulomatous amebic encephalitis (GAE) and amebic keratitis
 - *Balamuthia* - GAE

Naegleria fowleri

Distribution: worldwide

Habitat: Free living

Fresh water/lakes and ponds

Moist soil

Parasitic

Nasal cavity and CNS

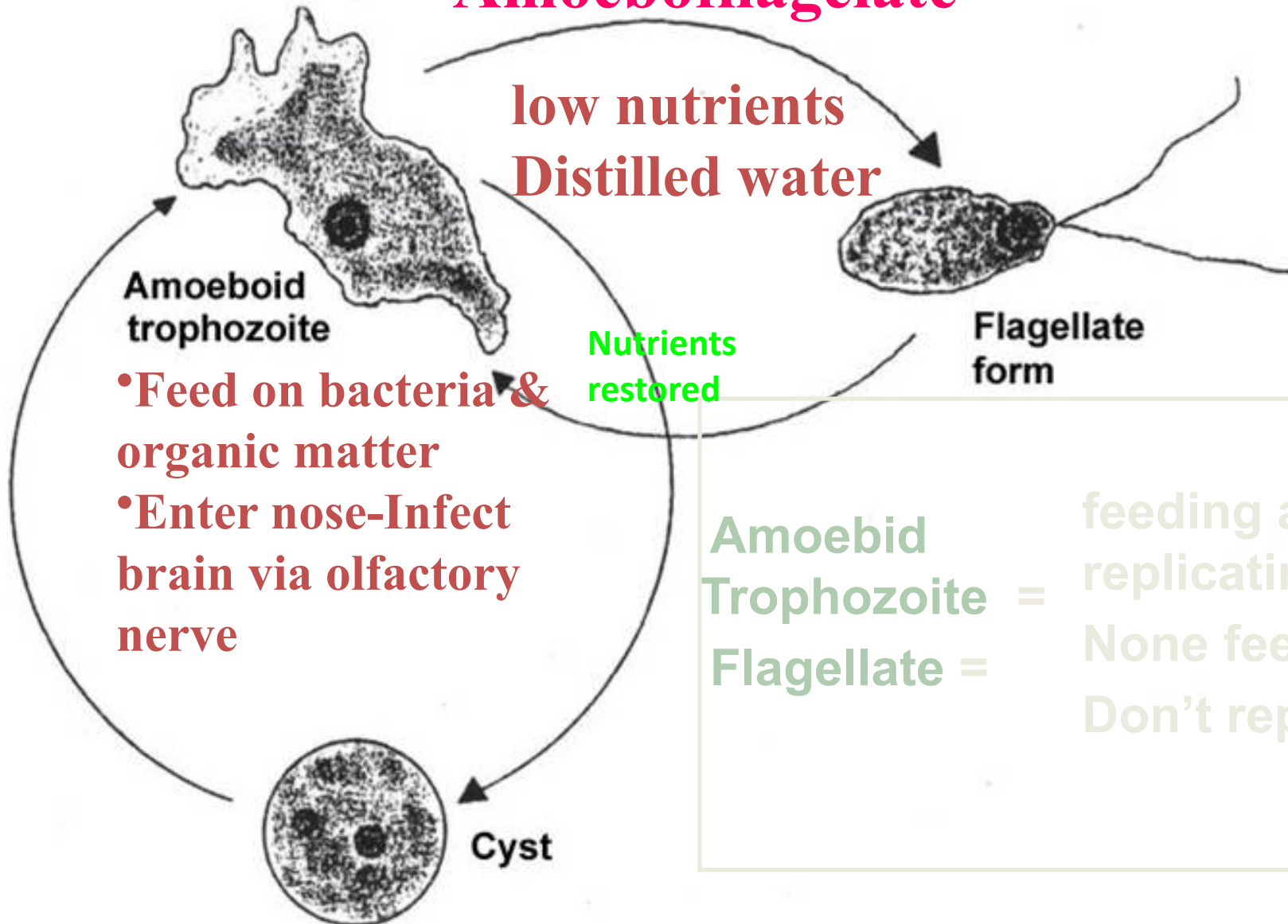
Causes: primary amebic meningoencephalitis
(PAM)

PAM first recognized by Fowler (1965)

~ 200 documented cases worldwide

Morphology & Life cycle

Amoeboflagellate



Amoebid
Trophozoite =
Flagellate =

feeding and
replicating form
None feeding
Don't replicate

Epidemiology

- *N. fowleri* has been isolated all over the world
- As of 1997, approximately 200 cases have been reported worldwide with 81 cases in the US (primarily in central and southeast)
- Risk factors: jumping/diving, IgA defici
- Males/females =3/1
- Mortality rate > 95%

Transmission

- **Infection:-** *through nasal cavity by aspiration of water contaminated with trophozoites while;-*
 - *bathing in stagnant fresh water/lakes/pools contaminated with sewage/decaying matter*
 - *under chlorinated swimming pool*
 - *Snuffing water from lakes/ponds etc*
- *Trophozoite invasive, reproductive form which thrives in temperatures from 35-46 0C.*

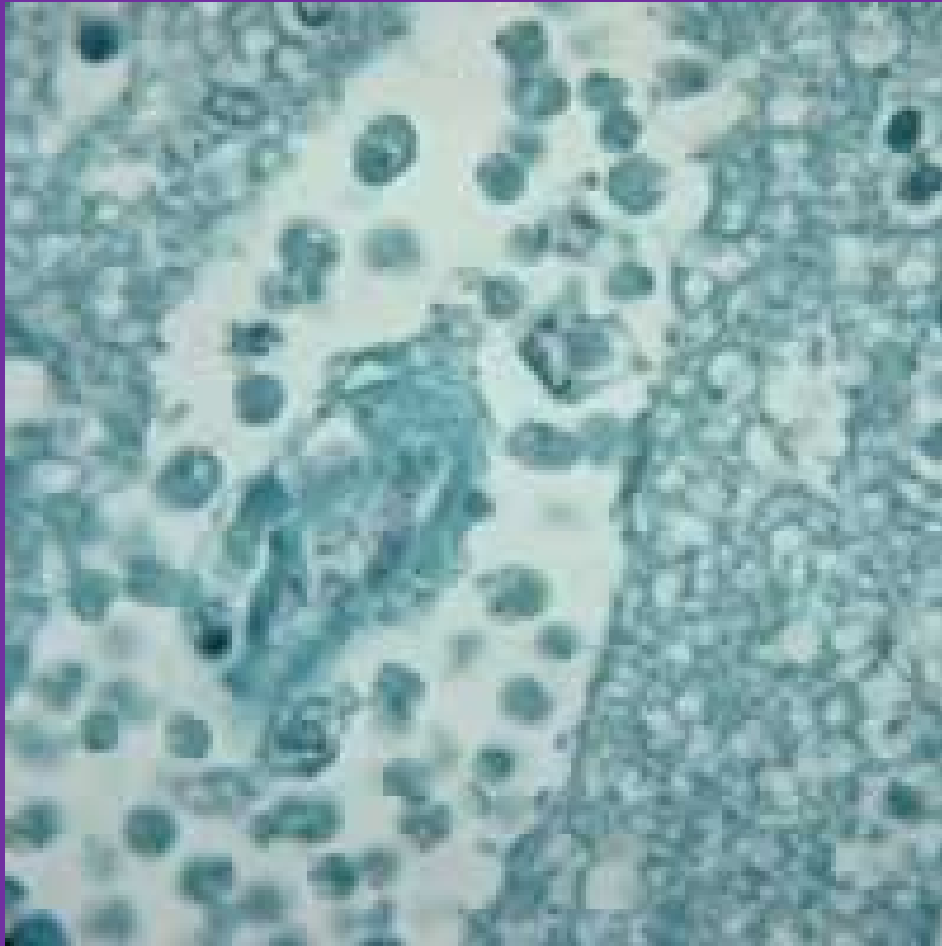
Pathophysiology

- Trophozoites penetrate the nasal mucosa and the cribriform plate during inhalation or aspiration of contaminated dust/water.
- The organisms migrate via the olfactory nerves to invade brain tissue
- Enzymes are produced resulting in diffuse and rapid hemorrhage and necrosis of the brain

pathology

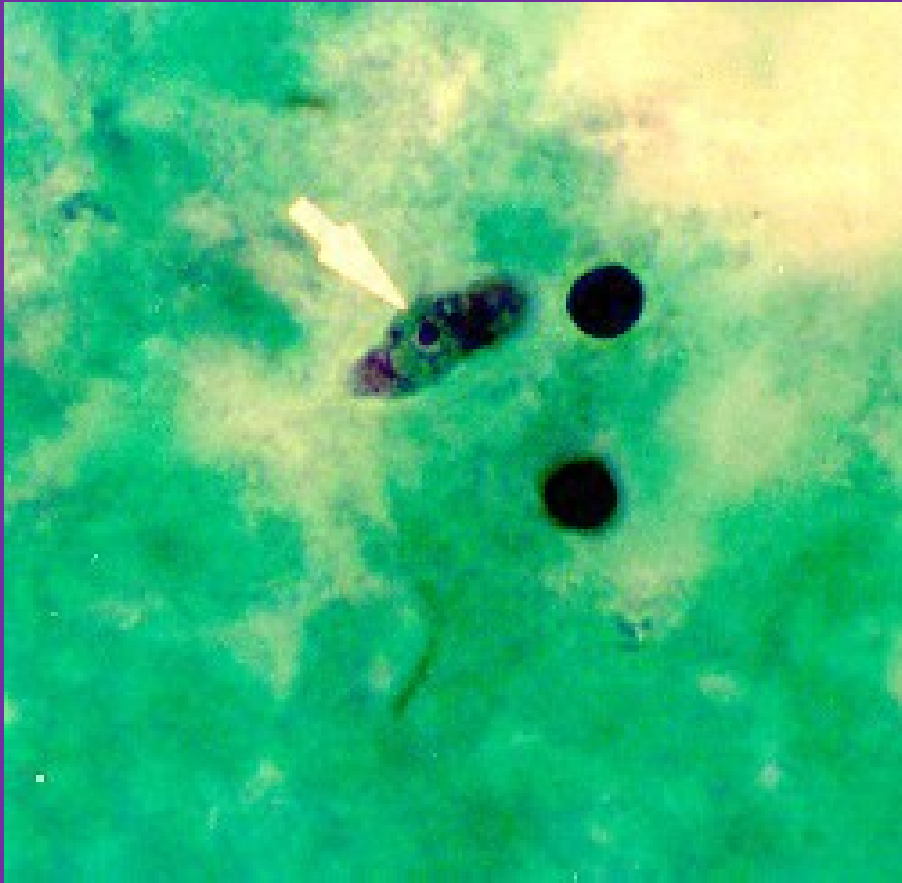
- **Causes Primary Amebic Meningoencephalitis (PAM)**
- Acute, infection of the brain and meninges caused by *N. Fowleri*
- symptoms usually within a few days after swimming in warm still waters(1-14 days)
- Initially may notice change in taste/smell
- symptoms include headache, lethargy, disorientation, coma
- Mental status changes seen in 66%
- rapid clinical course, death in 4-5 days after onset of symptoms

N. fowleri: Pathology



- Symptoms very similar to other types of meningitis and encephalitis.
- Indistinguishable from acute bacterial meningitis
- Very rapidly causes the death of host
 - Rapid destruction of brain tissue
- Usually affects immunocompetent children and young adults (range 8mo- 60years)

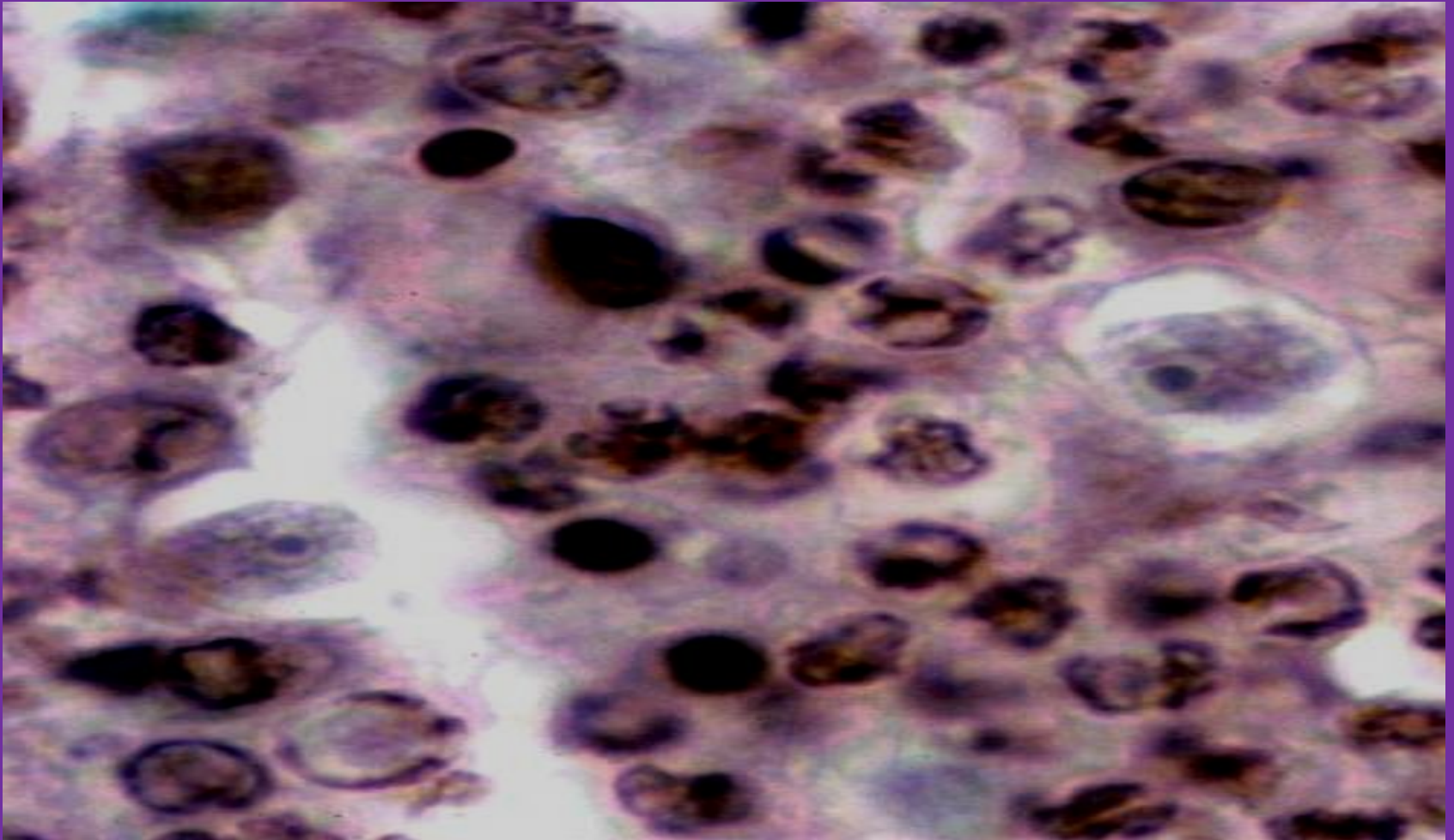
N. fowleri: Laboratory diagnosis



- Diagnosis is usually made at autopsy.
 - Amoeba in a brain smear
- Finding trophozoites in CSF

Indications of Naegleria infection: elevated white cell count in CSF without the successful recovery of bacteria

Masses of amoeba may be seen histopathologically in the brain



N. fowleri: Treatment

- Early diagnosis, drug therapy and supportive care are necessary in order to have ANY chance of survival (7 documented survivors)
- Treatment is rarely given
 - Amphotericin B has been used successfully in two human cases.
 - Very toxic
 - A traditional treatment is qinghaosu
 - A plant that also fights malaria

Prevention

- Keep head above water (or use nose plugs) in thermal pools or other warm, stagnant water.
- Avoid swimming or jumping into bodies of warm fresh water

Summary

- Consider *Naegleria* when all studies point to bacterial meningitis but the gram stain is negative
- Overall mortality is near 95% therefore you need a high index of suspicion
- Plug your nose or wear nose clips when jumping into fresh water



Acanthameoba sp.



Definitive Host: Usually
Free-living

- Facultative parasite of humans

Intermediate Host: None

Geographic Distribution:
Cosmopolitan

- Found in freshwater almost everywhere
 - Amoeba and cyst forms
- Cannot survive in thermal pools

Acanthameoba sp.

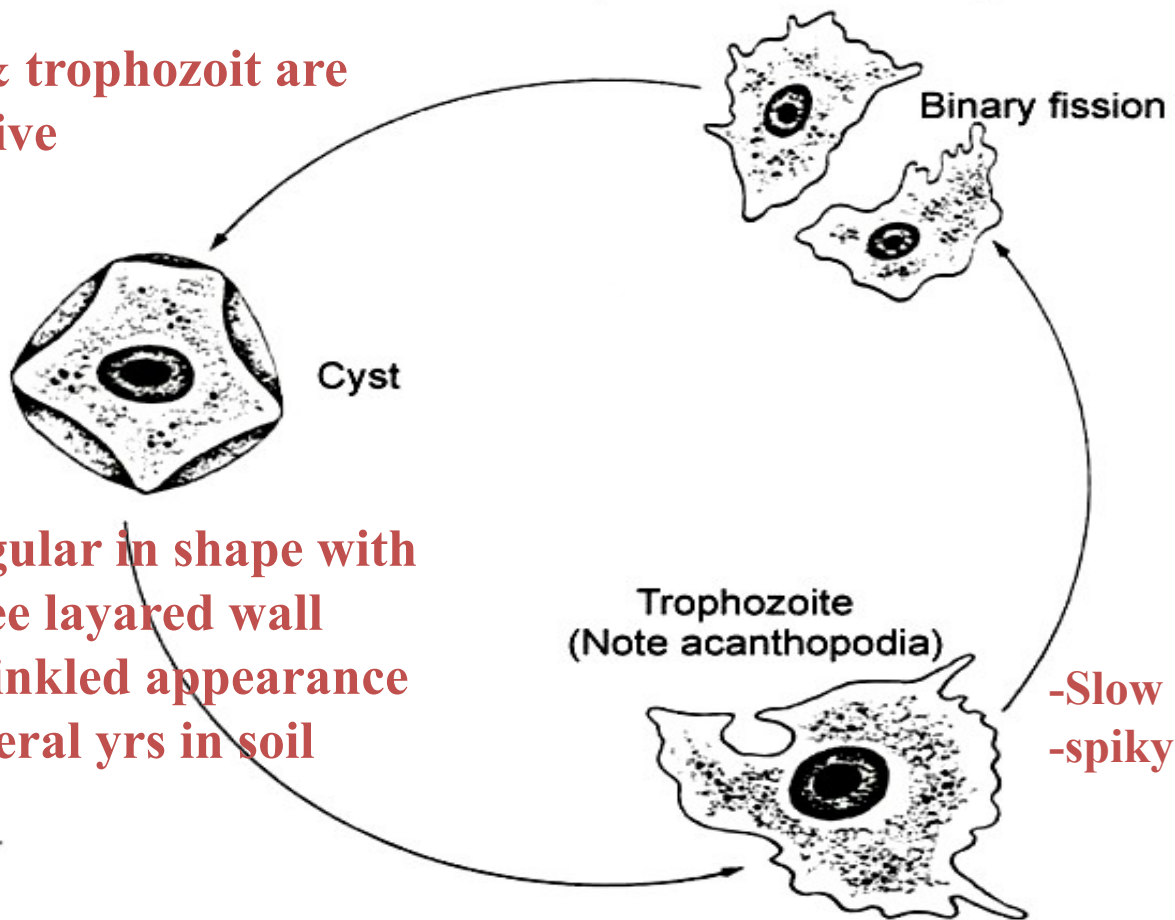
- **Mode of transmission:**
 - Cyst and trophozoites are infective
 - portal of entry unknown, possibly respiratory tract (inhalation of cysts)
 - invades body through cuts and abrasions or wounds in skin/eye contaminated with soil
 - The brain is probably infected by trophozoites via blood stream from infected skin or lung
- **Location in Host:** Most common in eye and skin. Rarely invades brain/lung

Morphology & life cycle

Typical protozoan life cycle

Exist as trophozoites and Cyst

cyst & trophozoite are
infective



Angular in shape with
three layered wall
Wrinkled appearance
Several yrs in soil

-Slow motility
-spiky projection

Acanthameoba sp: Pathology:

- Rarely causes damage in people with intact immune systems
- Most common cause of corneal ulcers and keratitis in contact lens wearers

Mainly causes

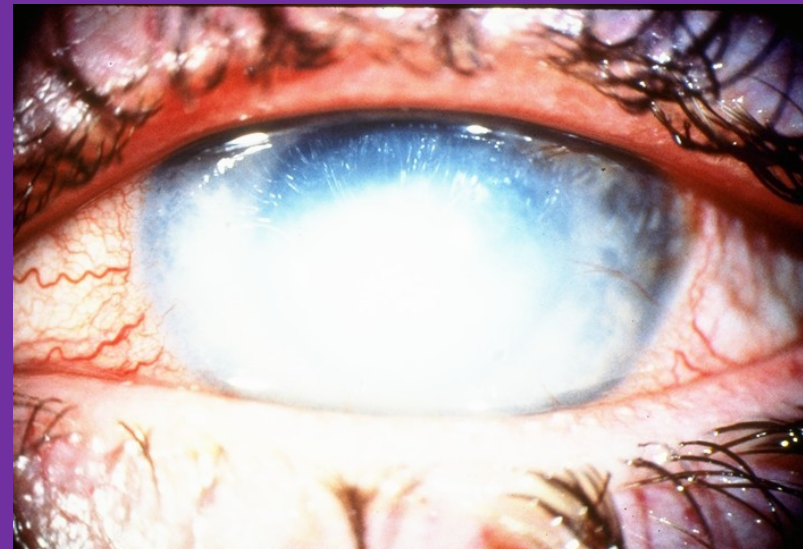
A- Amebic Keratitis

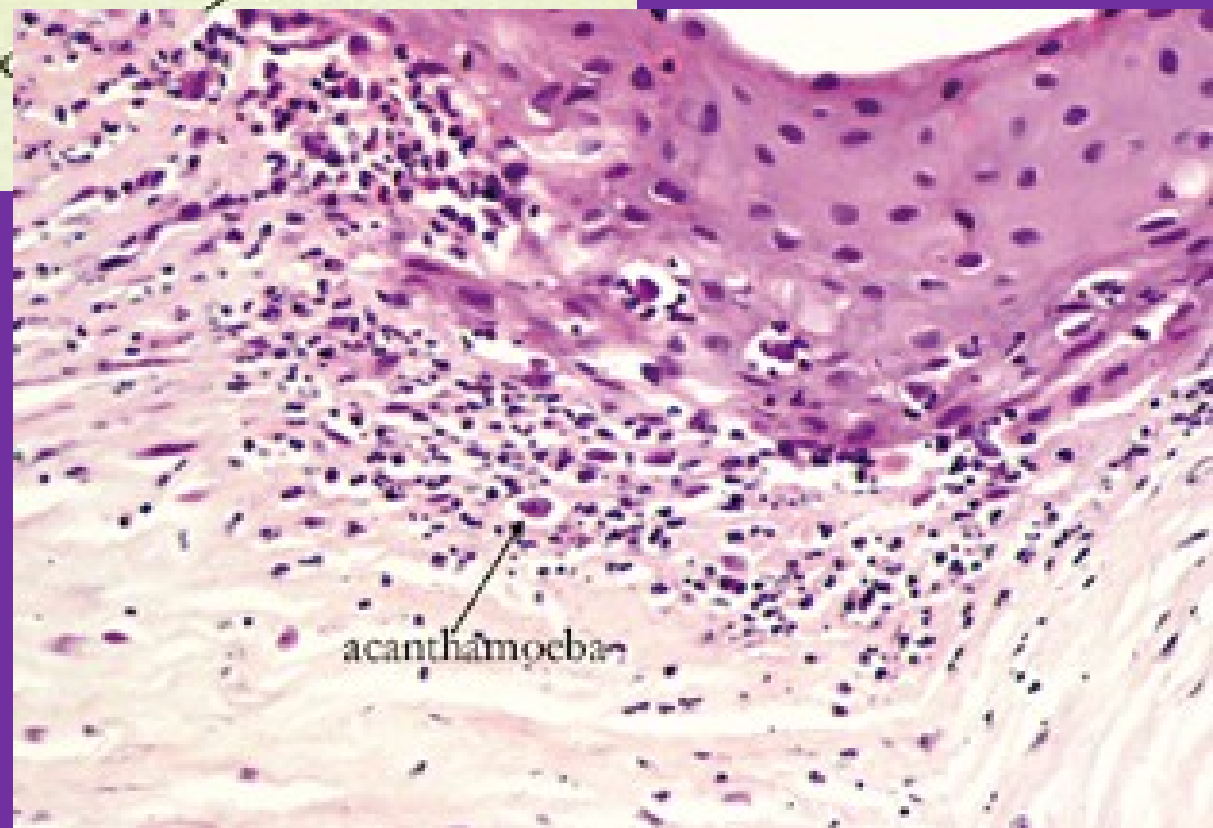
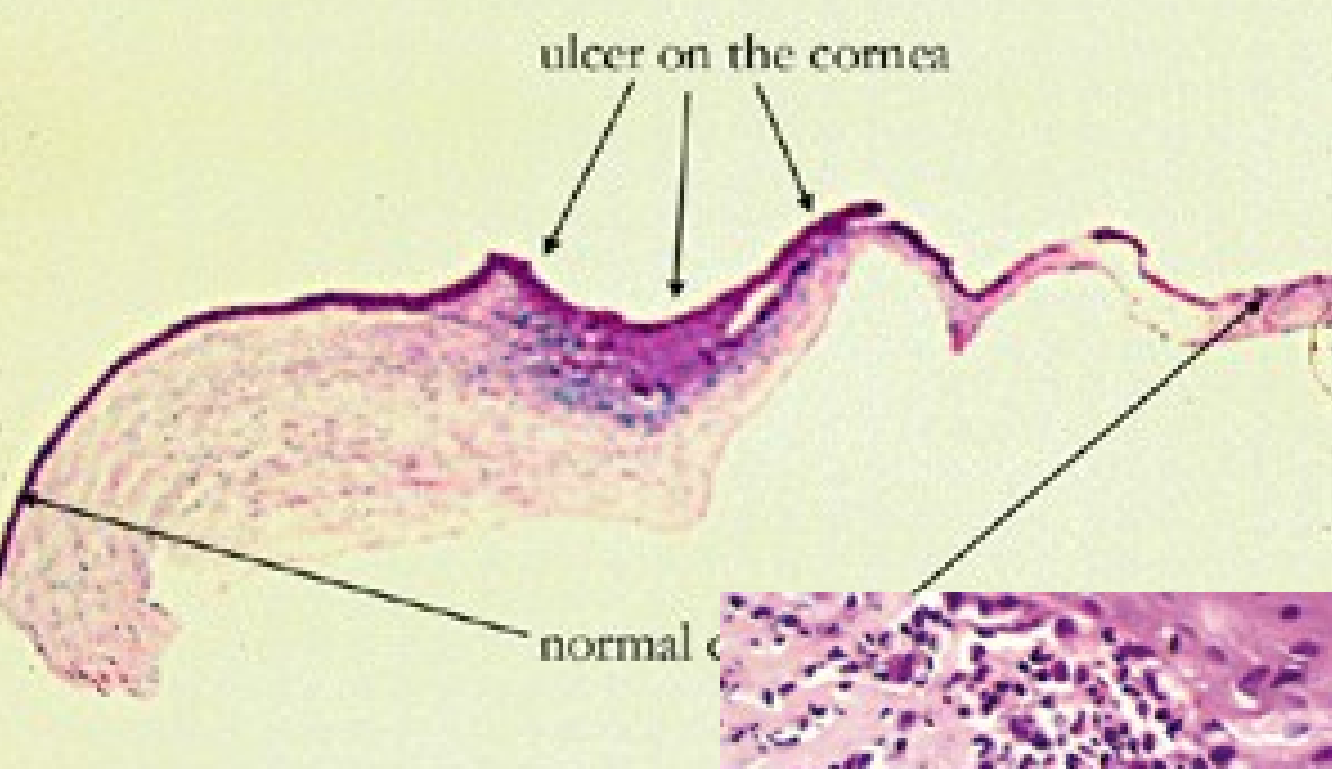
B- chronic Granulomates meningoencephalitis

B- Amebic

Keratitis

- A vision threatening chronic inflammation of the cornea
- predisposing factors
 - ocular trauma
 - contact lens
- symptoms
 - ocular pain
 - corneal lesions (refractory to usual treatments)
- diagnosis
 - demonstration of amebas in corneal scrapings
- Treatment- difficult, limited success





Acanthamoeba: Pathology

- AIDS Patients and other immune suppressed individuals cannot fight the amoeba
 - May cause skin ulcerations, keratitis, and corneal ulcerations
 - In rare cases, it can cause problems in the central nervous system of immune suppressed individuals.
 - Can cause meningoencephalitis like *N. fowleri*

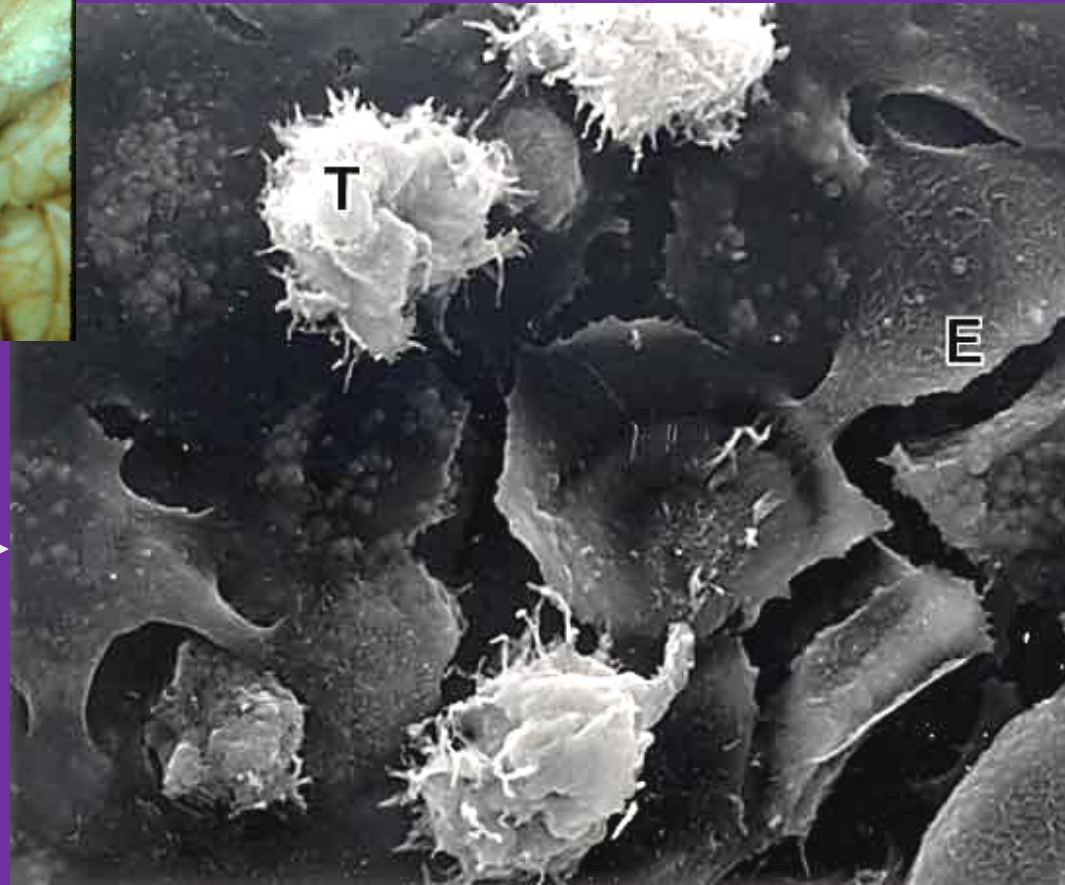
B- chronic Granulomates meningoencephalitis

- associated with immunosuppression
- onset is insidious with headache, personality changes, slight fever
- Prolonged clinical course, weeks to months to progresses to coma and death
- amebas not yet detected in spinal fluid
- In contrast to naegleria, both trophozoites and sometimes cysts are detected in histological examination
- no human cures documented
- Diagnosis is difficult usually done at autopsy



Brain damage due to
Acanthamoeba

Acanthamoeba killing
cornea epithelium cells



Acanthamoeba: Treatment

- Blindness has been treated by corneal transplants.
 - If corneal abrasion continues, stop wearing contacts.
- Treatment of people with suppressed immune system is difficult.
 - Miconazole and ketaconazole seem to work the best.
 - Some bacteria make chemicals that kill amoeba

Acanthamoeba: summary

- ubiquitous ameba of the soil and water
- human cases first reported in the early 1970's
- In contrast to PAM, it is associated with chronically ill, immunocompromised and other debilitated patient
- Like naegleria, is neutrotropic and causes encephelites
- GAE is slowly progressing and chronic type
- Also associated with lesion in the cornea

A comparison of Naegleria morphology with Acanthamoeba

Naegleria	Acanthamoeba
1. Trophozoite displays broad pseudopods 2. Activity motile 3. Form flagellate stage 4. Thin walled cysts 5. Cyst wall has no pores 6. Does not encyst in tissues	1. Trophozoite displays filamentous pseudopods (acanthopodia) 2. Sluggishly motile 3. Does not form flagellate stage 4. Double walled cysts 5. cyst wall may have pores or osteioles 6. May encyst in tissues

Giardiasis

- **Definition**
- A protozoan infection principally of the upper small intestine associated with symptoms of chronic diarrhea, steatorrhea, abdominal cramps, bloating, frequent loose and pale greasy stools, fatigue and weight loss.
- **Infectious agent**
Giardia lamblia
- **Epidemiology**
- **Occurrence-** Worldwide distribution. Children are more affected than adults. The disease is highly prevalent in areas of poor sanitation.
- **Reservoir-** Humans
- **Mode of transmission-** Person to person transmission occurs by hand to mouth transfer of cysts from feces of an infected individual especially in institutions and day care centers.

- **Period of communicability-** Entire period of infection, often months.
- **Susceptibility and resistance-**
- Asymptomatic carrier rate is high.
- Infection is frequently self-limited.
- Persons with AIDS may have more serious and prolonged infection.
- Recurrent infection is common among IgA deficient individuals

Life cycle

TRANSMISSION

1. Cysts ingested in food, water or from hands contaminated with feces.



HUMAN HOST

2. cysts excyst, forming trophozoites
3. Multiply in intestine
4. Trophozoites encyst.
5. Infective cysts passed in feces. *

* trophozoites passed in feces disintegrate.

ENVIRONMENT

6. Feces containing infective cysts contaminate the environment.



Clinical Manifestation

- Ranges from asymptomatic infection to severe failure to thrive and mal-absorption.
- Young children usually have diarrhea but abdominal distension and bloating are frequent.
- Adults have abdominal cramps, diarrhea, anorexia, nausea, malaise, bloating.

- **Diagnosis**

Demonstration of Giardia lamblia cyst or trophozoite in feces.

- **Treatment**

Metronidazole or Tinidazole

Giardiasis

Common

- **Manifest in 7-10 days**
- **Lasts 2- 6 weeks**
- **Symptoms**
 - Severe diarrhea
 - Abdominal cramps
 - Bloating
 - Flatulence
 - Nausea
 - Vomiting
- **May lead to**
 - Weight loss
 - Dehydration

vs.

Chronic

- **More severe condition**
- **Lasts 2 months - years**
- **Symptoms**
 - Increased gas
 - Dehydration
 - Burping
 - Loose stools
 - Slowed growth

- **Prevention and control**

1. Good personal hygiene, and handwashing before food

and following toilet use

2. Sanitary disposal of feces

3. Protection of public water supply from contamination of

feces

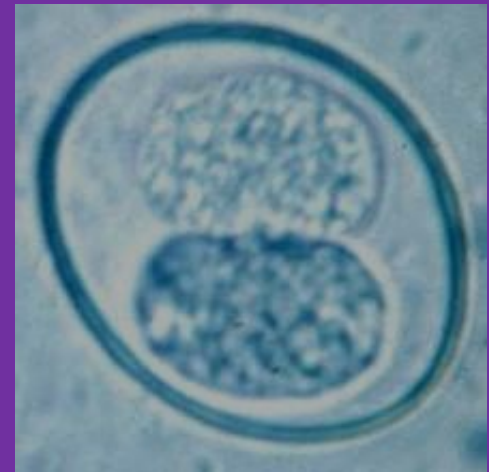
4. Case treatment

5. Safe water supply

Intestinal coccidian *Cryptosporidium* , *Isospora* , *Cyclospora*

General characteristics

- Considered as **opportunistic** parasite in immunocompromised person
- Complete entire **life cycle in single host**
- Within the **intestinal epithelial cells** of the host
- Characterized by a **thick walled oocyst** excreted in faeces
- Are transmitted by the fecal-oral route



Cryptosporidium

Species:

C. parvum

- A coccidian parasite implicated in intestinal
- Disease primarily among immunocompromised patient
- 1985 first reported case in philippines associated with intractable diarrhea in immunocompromised (AIDS) patient
- Self-limiting diarrhea in immunocompetent persons
- Profuse, watery diarrhea associated with AIDS (life threatening)

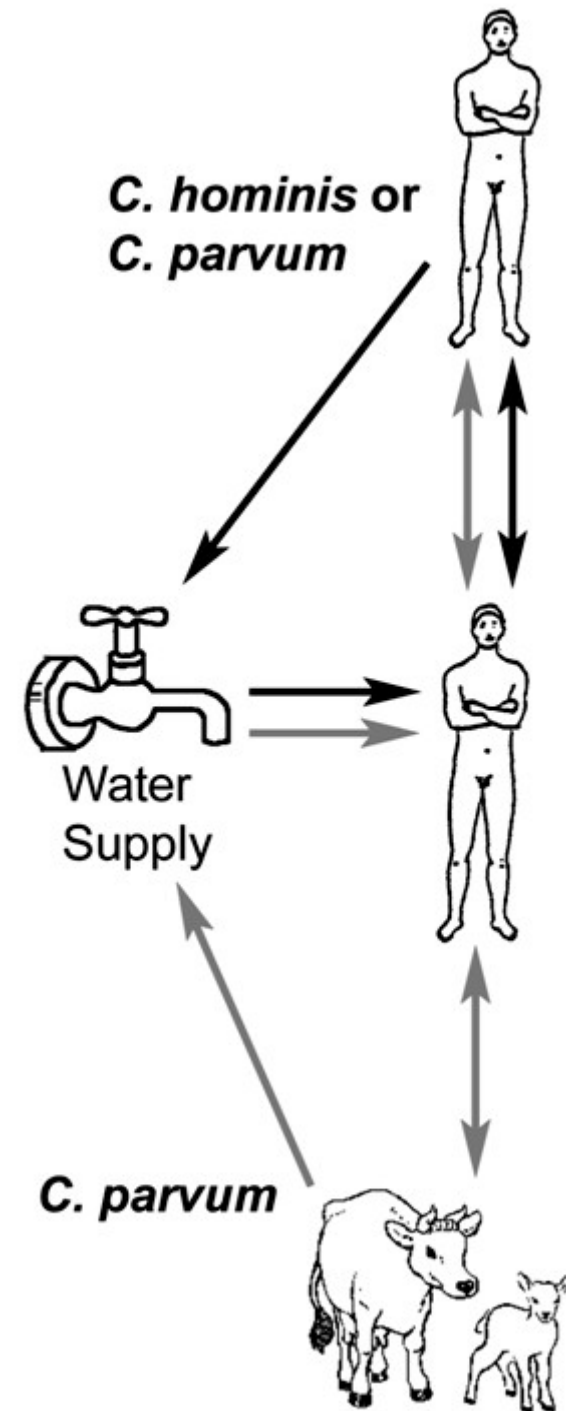
- 1976 considered as natural parasite of animal especially calves causing diarrheal diseases
- Infect wide range of animal hosts (*fish-humans*)
 - *C. hominis* & *C. parvum* is species mostly causing disease in humans
 - *C. hominis* infects only humans but *C. parvum* also infects many other mammals
 - *C. felis*, *C. meleagridis*, *C. canis*, and *C. muris* infections have also been reported

Transmission

- Mainly through the ingestion and possibly inhalation of sporulated oocysts
 - mainly through contaminated water &
 - Occasionally food sources, such as chicken salad
- fecal-oral transmission (monoxenous)
 - anthroponotic transmission
 - autoinfection
 - zoonotic transmission

Transmission

- 2 major types & sources:
 - *C. hominis*
 - anthroponotic transmission
 - only human sources
 - non-infective for mice or calves
 - *C. parvum*
 - human and bovine sources
 - infective for mice and calves
 - zoonotic & anthroponotic transmission
 - waterborne outbreaks is common
 - in Milwaukee (Wisconsin) in 1993, affected > 400,000 people



Waterborne Cryptosporidiosis

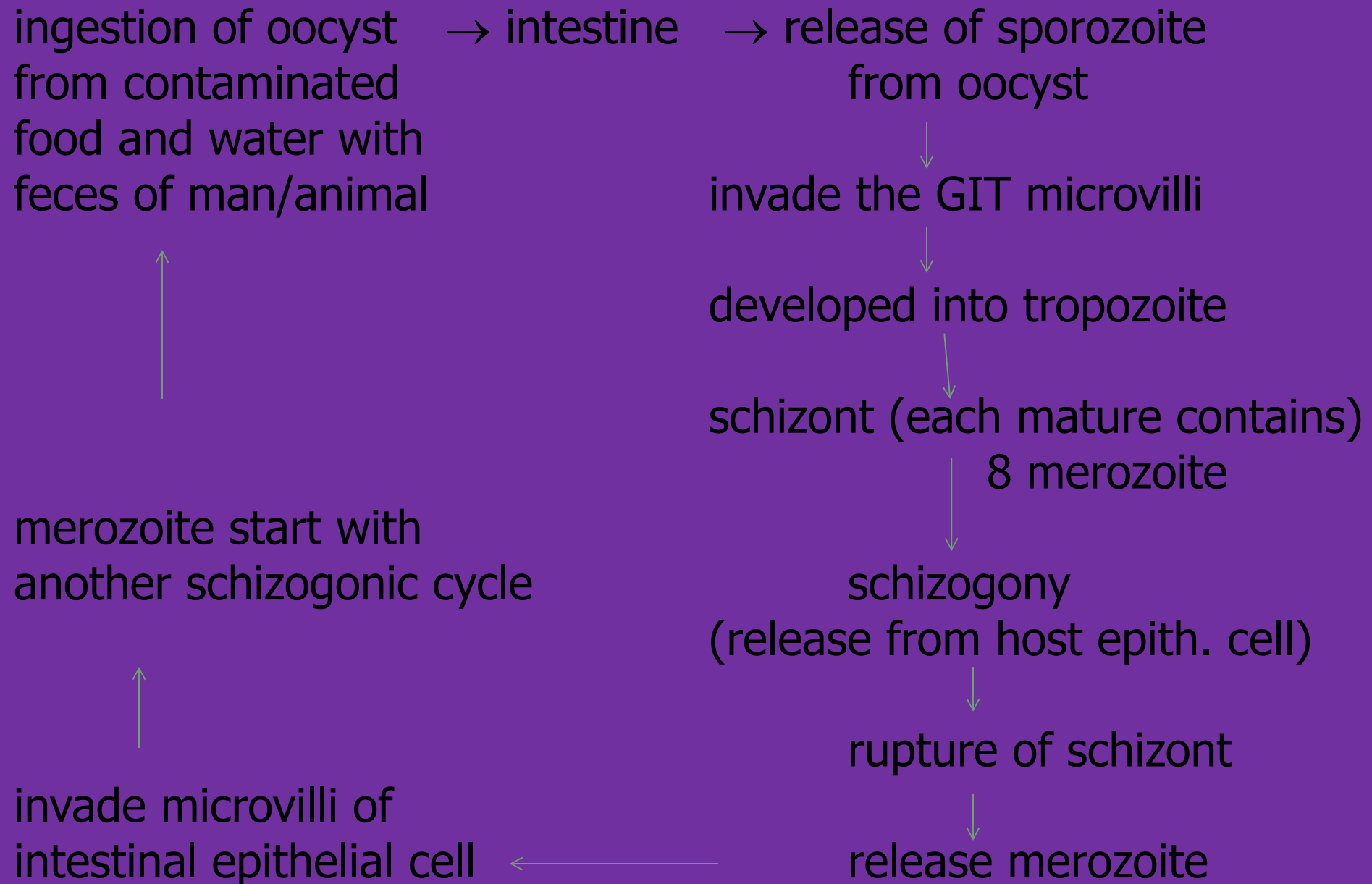
- human infection usually waterborne and acquired by fecal-oral route
- highest prevalence of disease in areas with unreliable water and food sanitation

Factors Favoring Waterborne Cryptosporidiosis

- Small size of oocysts (4-5 μ m)
- Reduced host specificity and monoxenous development
- Close associations between human and animal hosts
- Large number of oocysts excreted (up to 100 billion per calf per day)
- Low infective dose (<30)
- Robust oocysts; resistant to chlorine

Life Cycle:

Asexual:

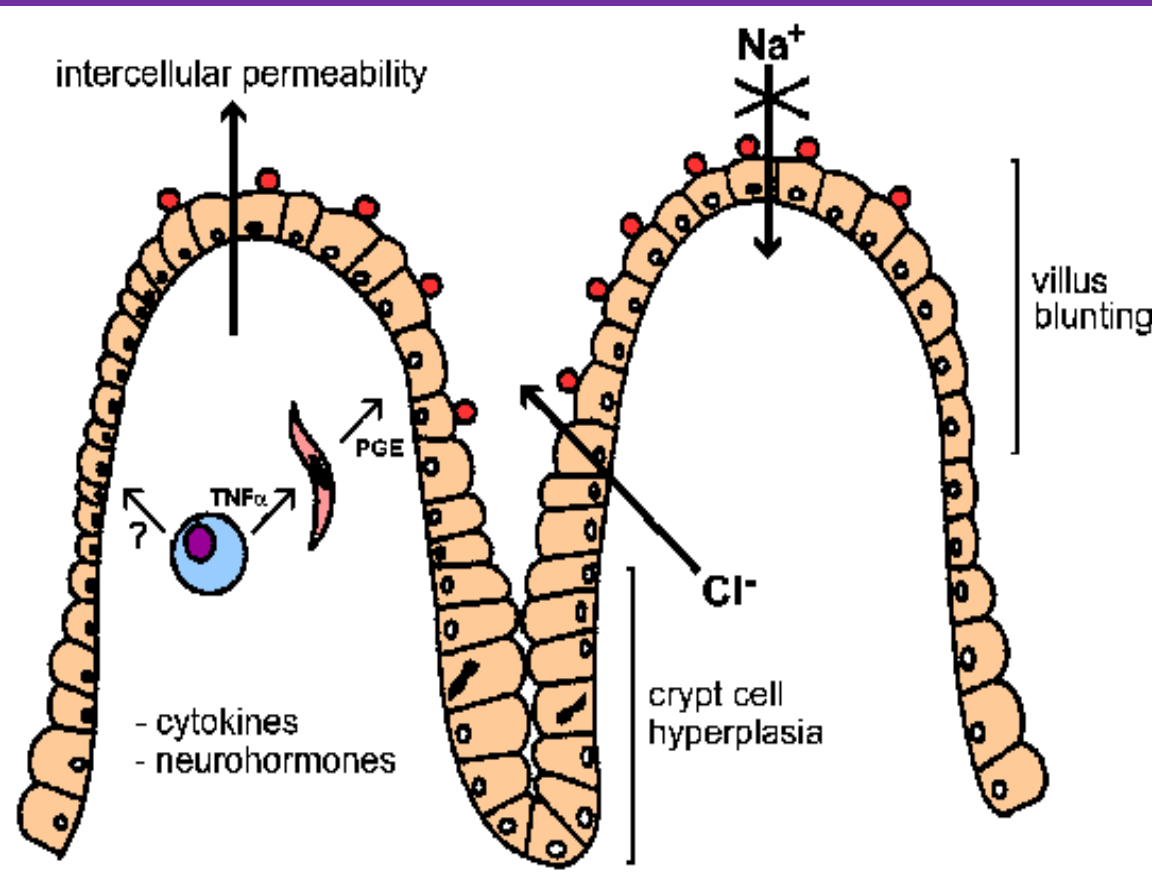


Sexual: = Gametogenesis and Fertilization of male and female gametocytes → zygote → development of oocyst → sporogony → 4 sporozoites with in oocyst → passed out in feces

Pathogenesis

- Enterocytes damaged or killed
 - Villus atrophy (blunting)
 - Enterocyte malfunction
 - Impaired absorption
 - Enhanced secretion
- Crypt cell hyperplasia
- Inflammation in lamina propria
 - Impaired absorption
 - Promote secretion

Pathogenesis



Schematic representation of *Cryptosporidium* pathogenesis. Many factors may be involved in the diarrhea associated with cryptosporidiosis. (Modified from Clark and Sears, *Parasitology Today* 12:221, 1996.)

- **Osmotic (primary)**
 - characterized by enterocyte malfunction
 - $\downarrow$ Na⁺ absorption
 - $\uparrow$ Cl⁻ secretion
- **Inflammatory**
 - generally associated with invasion of the mucosa
 - inflammation of the lamina propria
 - leukocytes in the stools
- **Secretory**
 - generally associated with bacterial enterotoxins
 - characterized by watery diarrhea
- Increased intercellular permeability and inflammation in the submucosal layer

Disease: Cryptosporidiosis

- = human infection usually waterborne and acquired by fecal-oral route
- = highest prevalence of disease in areas with unreliable water and food sanitation
- = extraintestinal infection of the respiratory tract, biliary tract and pancreas may occur.

Clinical Features

- Vary from asymptomatic to severe, life-threatening illness;
- infection usually self-limiting diarrhea of 1-2 weeks duration characterized by copious watery diarrhea, vomiting, intense abdominal pain, anorexia and weakness
- among immunocompromised patient (AIDS) → develops severe chronic diarrhea which may last for months → malabsorption

Clinical Features....

- Incubation period ~ 7 days (2 to 10 days).
- In immunocompetent persons,
 - Symptoms are usually short lived (1 to 2 weeks)
- Persons with AIDS (CD4 counts $<200/\mu\text{l}$)
 - Chronic and more severe lasting for months or even years
- Small intestine- most commonly affected
 - the lungs, and possibly conjunctiva may be affected.

Cyclospora

Specie: *C. cayetanensis*

- = was established to cause human diarrhea in 1990
- = 1986 cases of prolonged watery diarrhea among immunocompromised (AIDS) patient has been reported worldwide
- = Species Name C. *Cayetanensis* was given in 1993
 - = Initially called 'cyano-bacteria like body' (CLB) or large *cryptosporidium*

Epidemiology:

- = More common in tropical and sub-tropical areas
- = 1979 first case of cyclospora infection was reported in papua, new guinea
- = Subsequent cases has been reported from most part of the world

Epidemiology...

- = Infection caused by cyclospora can be acquired by drinking contaminated water (fecal-oral)
- = More associated with food-borne outbreaks(social events, weddings, etc)
- = Parasite infect vertebrates including reptiles, insect & rodents
 - Life cycle similar to cryptosporidium except that Oocysts mature in environment (days-weeks)

Clinical Features

Disease: Cyclosporiasis

- = disease is clinically indistinguishable from cryptosporidiosis and Isosporiasis
- = self-limiting, characterized by persistent watery diarrhea that ends to recur in a relapsing pattern and last for 3-4 wks,
- = associated with abdominal cramps, nausea, vomiting, low grade fever, weight loss and anorexia

Pathogenesis:

- Infection typically confined to the jejunum
- characterized by inflammatory changes, villous atrophy and hyperplasia of the jejunal tissue

Isospora

Specie: Isospora belli

Epidemioogy

- wide geographical distribution (higher prevalence in warmer climates)
- the least common of the three intestinal coccidia that infect humans
- transmitted fecally in contaminated food and drink with oocyst

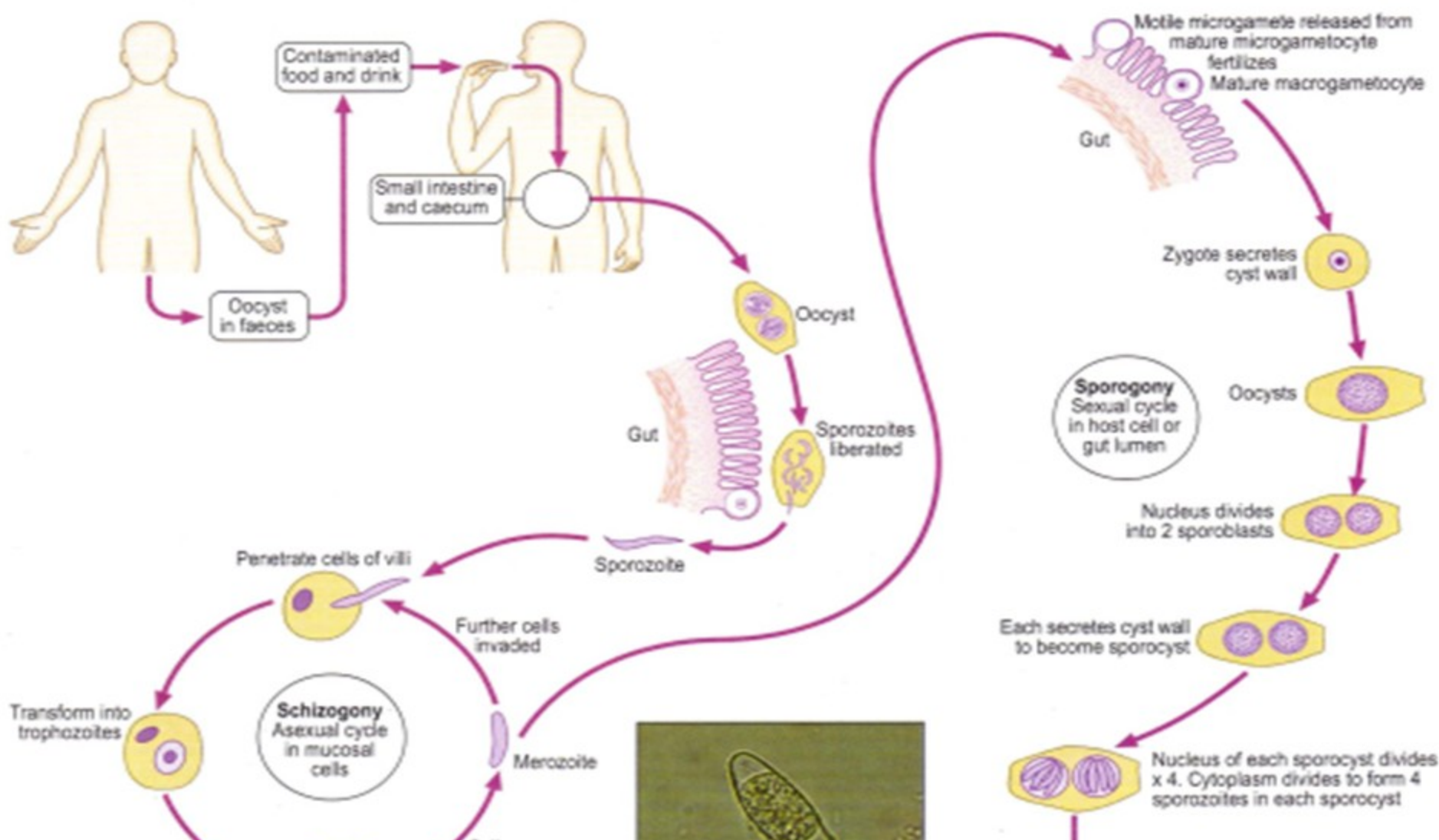
Habitat:

intestinal tract probably in the ileum and cecum

life cycle similar to *Cryptosporidium* except that oocysts mature in environment

Isospora belli (causing coccidiosis in humans)

Life cycle



Clinical feature

- **Disease:** Isosporosis/Intestinal coccidiosis
- often asymptomatic (seldom reported)
- symptoms range from mild gastro-intestinal distress to severe dysentery
- mild self-limiting infection charac. by fever, abdominal pain, severe diarrhea, steatorrhea (fatty stool) and weight loss.
- often self-limiting, but can become chronic (wasting, anorexia)
- symptoms more severe in AIDS patients

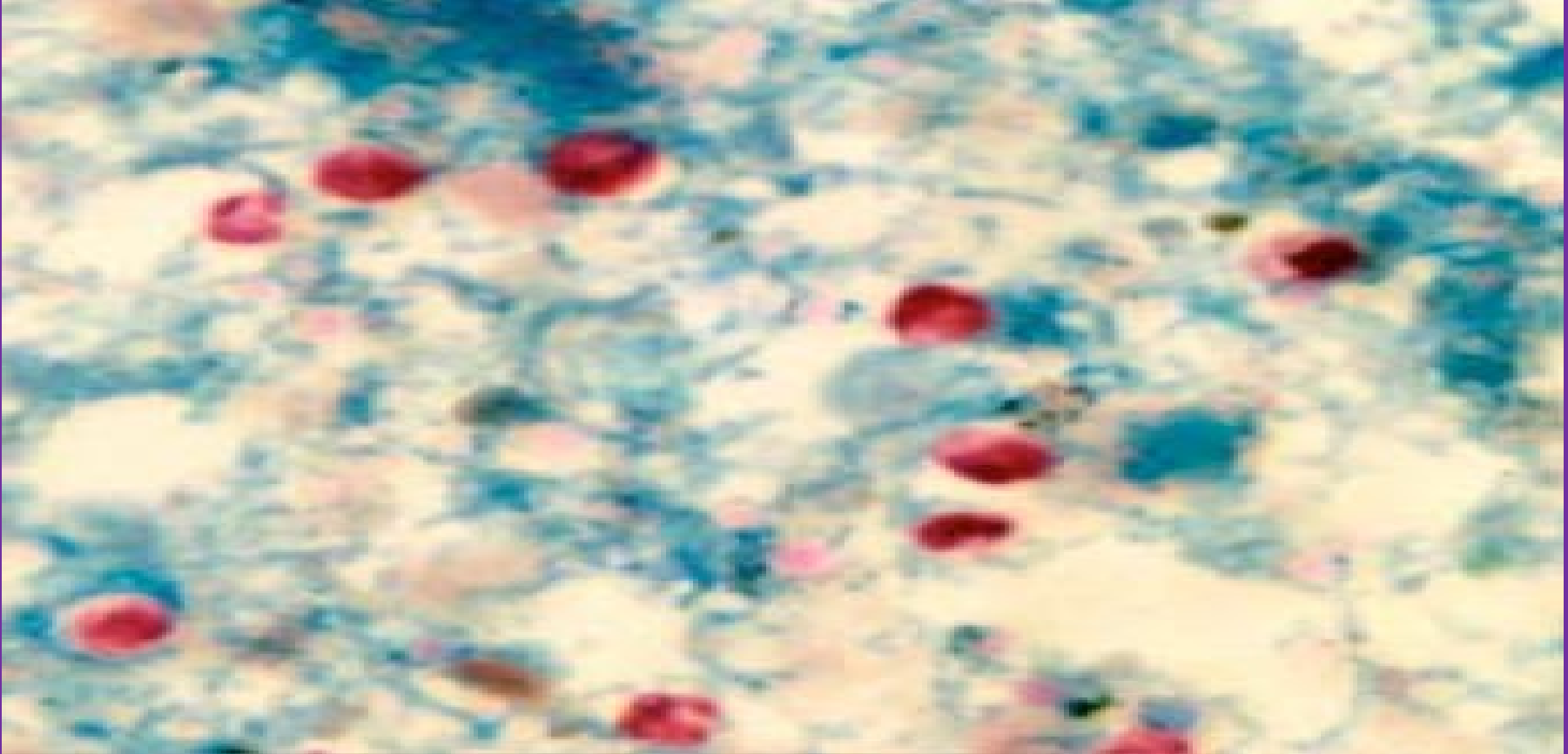
Pathology

- intracellular development occurs in the intestinal mucosa (lamina propria)→ mucosal atrophy
- charac. chronic diarrhea (months to years) abdominal discomfort, low grade fever

Laboratory diagnosis Intestinal coccidian

- Diagnosed by finding oocysts in faecal specimens
- Specimens are usually watery and often have an offensive smell. Pus cells are not found.
- Oocyst of intestinal coccidian can be detected in wet preparations but they are more easily identified in smears stained by modified Ziehl-Nelson/ Acid fast staining.

acid fast-stained oocysts of *Cryptosporidium parvum*



Treatment Intestinal coccidian

- *I. belli* and *C. caytenensis* can be treated using trimethoprim- sulfamethoxazole
- There is no effective treatment for *C. parvum* but paromomycin
 - modest benefit
 - lowers parasitemia in AIDS
- prevention and control
 - Safe water supply
 - Personal and environmental hygiene
 - Health education
 - Treatment

Balantidiasis (Balantidial dysentery, Ciliary dysentery)

- Caused by protozoan *Blantidium coli*

Epidemiology

World wide being more commonly found amongst those who keep pigs, and uses pig faeces as fertilizer especially in warmer climates

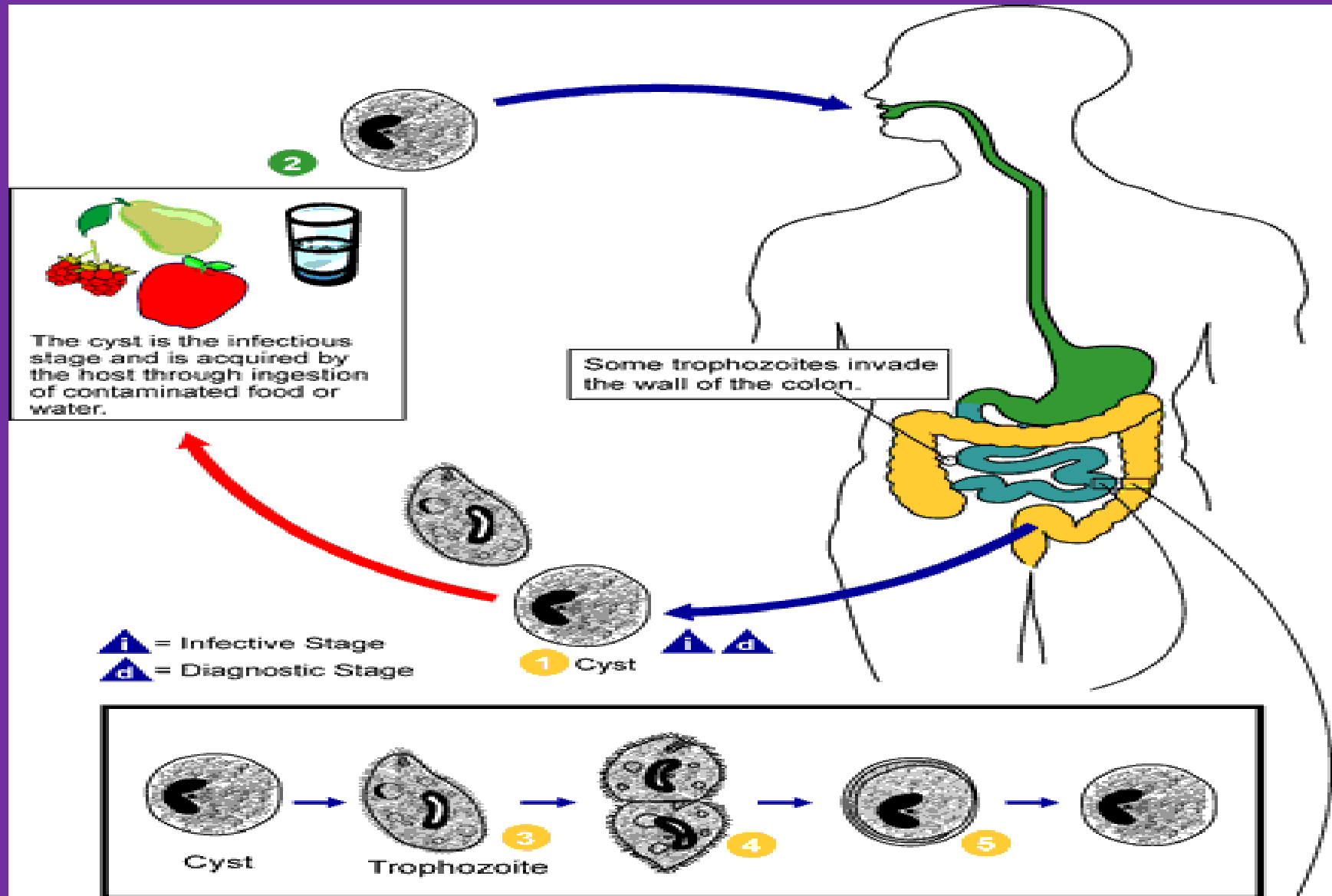
- In Ethiopia it was reported from Debre Brerahn.

Habitat: Trophozoite and cyst in the large intestine of pig and rarely man

Transmission and life cycle

➤ **Human** acquires through ingestion of contaminated food or water or hands contaminated with pig faeces containing cyst

life cycle



- Following ingestion, excystation occurs in the small intestine
- The trophozoites reside in the lumen of the large intestine of humans and animals, where they replicate
- Trophozoites undergo encystation to produce infective cysts
 - Some trophozoites invade the wall of the colon and multiply
 - Some return to lumen and disintegrate
- Mature cysts are passed with feces

Clinical feature

- Infection with *B.coli* can be without symptoms unless the ciliates invade the intestinal wall
- Invasion can cause inflammation and ulceration, leading to dysentery with blood and mucus being passed with faeces

Laboratory Diagnosis

- Finding the **tropozoites** in dysenteric faecal specimens and the **cysts** in formed or semiformed faeces
- In dysenteric specimens the ciliates usually contain ingested red cell

Treatment

- Tetracycline, metronidazole

Prevention and Control

- 1.Avoid contamination of food or drink
- 2.Improving personal hygiene especially those who keep pigs.
- 3.Treatment and health education.

Toxoplasma gondii

- Causative agent of toxoplasmosis
- Tissue or extraintestinal coccidia
- infects most species of warm blooded animals
, including humans

Epidemiology

- World wide

- About 47% of African , more than 50% adult population in north America and west Europe shows antibodies to *T. gondii*
- Only about 1% of persons showing antibody to *T. gondii* have signs and symptoms of diseases
- **In Ethiopia** prevalence of toxoplasmosis is > 80%

Habitat

- In the **skeletal muscle, heart, lymph nodes, lungs, spleen, bone marrow, mononuclear leukocytes, brain, CSF, Spleen, etc.** of man, domestic and wild animals

Morphology

- has five main developmental forms
 - ✓ only **trophozoite** and **cyst** stages are found in man
 - ✓ but all occur in the feline (cats family)

Toxoplasma(trophozoite):-has two forms

I. Tachyzoite/endozoite

- ✓ occurs in the early acute stage of infection

2. Bradyzoites/cryptozoites

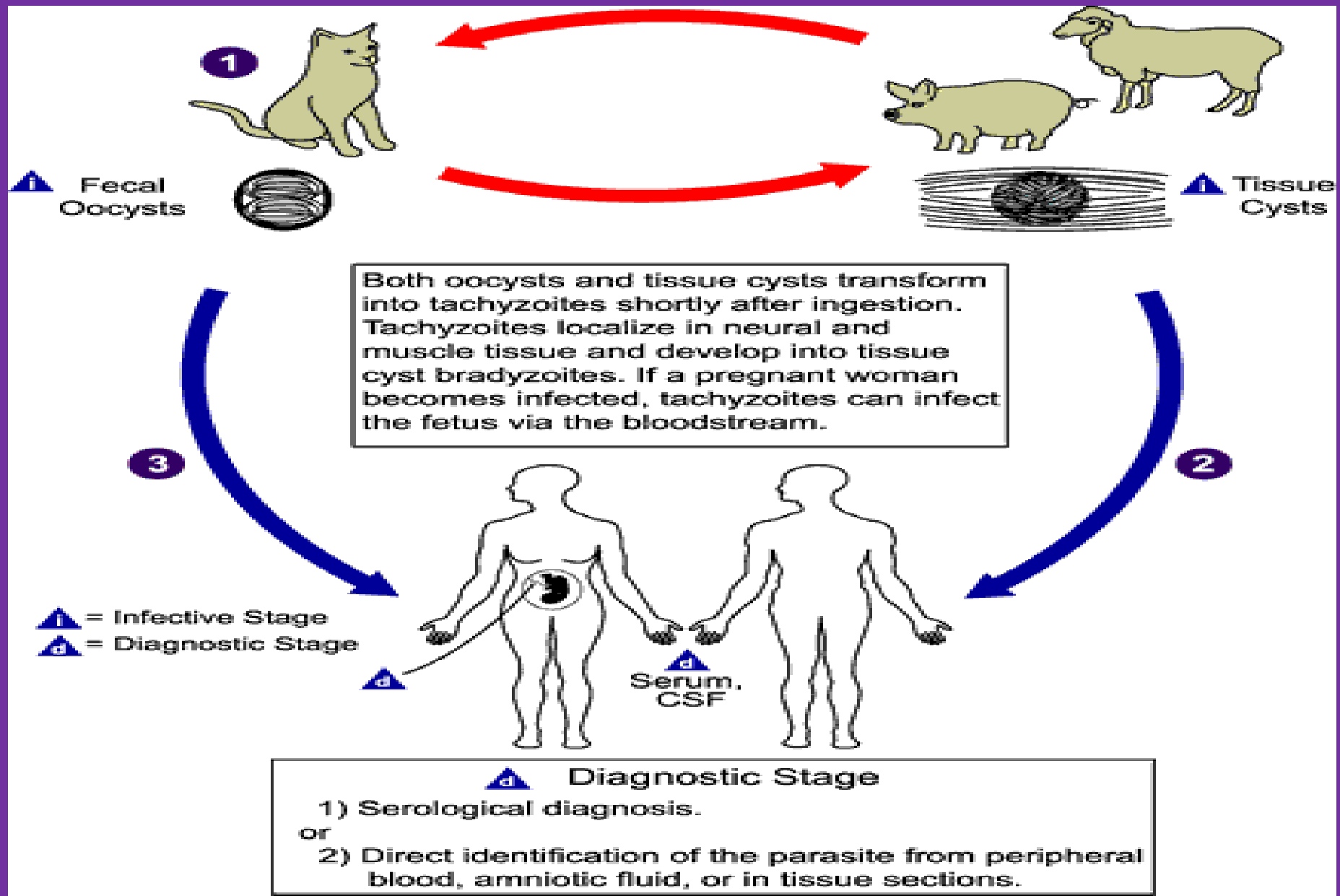
- Occurs in the chronic stage of infection, develops slowly and multiplies in the tissues to form a true cyst

Cyst:-10-100 μ m & may contain about 3,000 trophozoites

Transmission and life cycle

- Human infection may be acquired by
 - A) ingestion of undercooked infected meat containing *Toxoplasma* cysts
 - B) ingestion of the oocyst from fecally contaminated hands or food
 - C) organ transplantation or blood transfusion;
 - D) transplacental transmission;
 - E) accidental inoculation of tachyzoites

life cycle



Clinical Features

- adults acquired infection are often asymptomatic , they can cause
 - fever, rash
 - enlargement of lymph gland with lymphocytosis
 - occasionally inflammation of the eye (ocular toxoplasmosis) , myocarditis , meningoencephalitis and atypical pneumonia.

- **HIV associated toxoplasmosis**
 - serious & often fatal opportunistic toxoplasma infections
 - infections are due to the **reactivation of cysts**
 - cause encephalitis with fever , headache, mental deterioration and seizures

- **Congenital infections**

- occur in about 1-5 per 1000 pregnancies of which -
 - 5-10% result in miscarriage and
 - 8-10% result in serious brain and eye damage to the fetus
- 10-13% of the babies will have visual handicaps
- Infection can be acquired due to active parasitaemia during pregnancy or reactivated in immuno compromised mothers

Laboratory Diagnosis

1. Identifying toxoplasma in Giemsa stained histological sections, aspirates of lymphnode, bone-marrow, CSF, pleural fluid, peritoneal fluids and sputum
2. Serologic tests such as Sabin-feldman dye test, ELISA, IFAT, CFT.

Treatment

- ✓ Pyrimethamine-sulfadiazine (Fansider) or Spiramycin

Prevention and Control

1. Avoid contamination of hand, food and water with the faeces of cat
2. Not eating raw or under cooked meat such as pork, beef THANK YOU!

Cryptosporidiosis

- Protozoa *Cryptosporidium parvum* causes the disease Cryptosporidiosis.
- **Reservoir:**
 - Humans
 - Cattle
 - Other domestic animals
- It is a zoonotic disease and can travel from animals to humans.
- *Cryptosporidium parvum* has been recognized as a human pathogen since 1976.

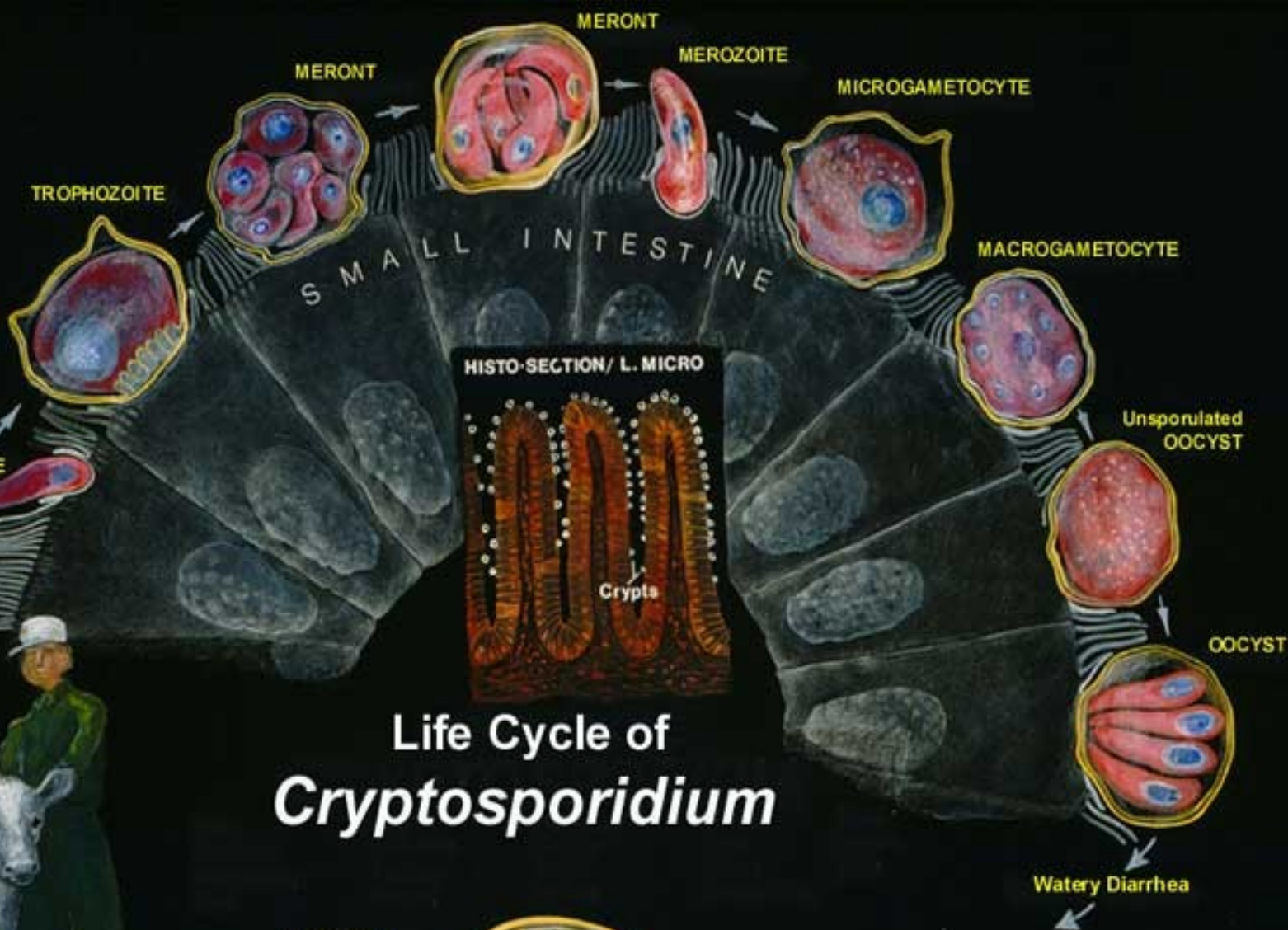
Site of infection

- Epithelial cells of the gastrointestinal tract
- Has affected other tissues such as respiratory tract tissues and conjunctiva of the eye.
- Infectious dose: < 10 organisms (only 1 needed to initiate)
- Cell death is a direct result of parasite invasion, multiplication, and extrusion **or**
- Cell damage could occur through T cell-mediated inflammation, producing microvilli death and *Cryptosporidium* excess growth

Cryptosporidium parvum

- Enteric parasite
- Invade and grow intracellular in mucosal epithelial cells of small intestine
- One of the three most common diarrhea-causing pathogens in the world

Invade and grow intracellularly in mucosal epithelial cells
of the stomach and intestine



Life Cycle of *Cryptosporidium*



Contamination of the environment



Mode of Transmission

- Fecal-oral
- Person to person
 - esp. child care settings
- Waterborne
 - RWI's, both natural and swimming pools
 - Drinking water
- Foodborne